Test-Time Adaptation of an Offline Multimodal Foundation Model for Simultaneous Speech Translation

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Abstract

End-to-end simultaneous speech-to-text translation (SimulST) systems typically rely on complex architectures and sophisticated training strategies. In contrast, we propose a simple approach that combines conventional pause-based segmentation for streaming audio input with a strong off-the-shelf multimodal foundation model *adapted at test-time* for translation. To achieve simultaneity, we adopt a variant of the classic wait-k read-write policy to control the interaction between audio input and translation output, and use a multi-turn conversation format with response prefilling and key-value caching for coherent translation and computational efficiency. Experiments on the official development sets of the IWSLT 2026 SimulST shared task show that our system achieves a better quality–latency trade-off than the cascaded baseline across all language directions and latency regimes, highlighting the effectiveness of this simple yet powerful approach.

1 Introduction

Simultaneous speech translation (SimulST) aims to translate source-language speech into target-language text in real time, minimizing delay to enhance user comprehension (Papi et al., 2025). SimulST is challenging due to the inherent quality–latency trade-off and limited availability of task-specific training data (Ko et al., 2023).

Conventional end-to-end SimulST systems are generally divided into two modules: one to handle audio segmentation and the other to determine when to read or write (Liu et al., 2024). These systems typically employ sophisticated adaptive segmentation and/or read-write strategies and architectures as well as specialised training techniques (Liu et al., 2024). An example of using an adaptive strategy for performing audio segmentation is Tsiamas et al. (2022), whereas Dong et al. (2022), Zhang and Feng (2023) and Zhang et al. (2022) are representative cases of learning an adaptive segmentation module. Similarly, adaptive read-write policies were explored by studies like Ma et al. (2020), Polák et al. (2023), Papi et al. (2023), and Ouyang et al. (2025). Despite their effectiveness, these methods typically entail considerable architectural complexity and training cost.

Motivated by the need for practical and efficient SimulST systems, we participate in the speech-to-text track without extra context of the IWSLT 2026 SimulST shared task, and propose a simple yet effective end-to-end SimulST system based on an offline multimodal foundation model. Inspired by Papi et al. (2022) and Deng et al. (2025), our system uses conventional pause-based audio segmentation to process streaming input and an instruction-tuned multimodal foundation model to perform translation. To enable an offline model to operate in a simultaneous setting at test time, we adopt a variant of the classic wait-k read-write policy (Ma et al., 2019). Following Ouyang et al. (2025), we formulate the translation process as a multi-turn conversation, in which each user message contains a new audio chunk and each assistant response provides the accumulated translation with response prefilling for cross-turn coherence. Our contributions are summarised as follows:

- We propose an end-to-end SimulST framework based on pause-based speech segmentation and a multimodal foundation model.
- We adapt the foundation model to simultaneous translation using a deterministic wait-k policy and a multi-turn conversation format.
- We use response prefilling and key-value caching to improve coherence and computational efficiency.
- We achieve a better quality–latency trade-off than the cascaded baseline across multiple language directions without fine-tuning or specialised training for simultaneous translation.

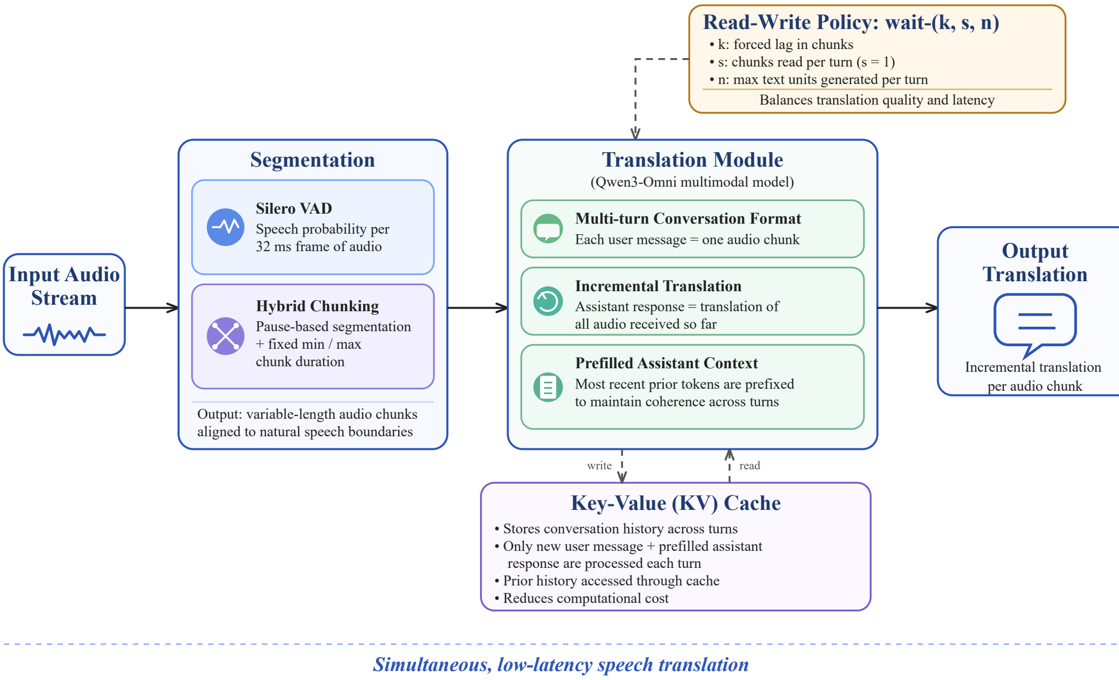

Figure 1: Overview of the proposed SimulST system via test-time adaptation of a multimodal foundation model.

2 System Architecture

As illustrated in Figure 1, the proposed system consists of two main modules: one for speech segmentation and the other for speech translation. The segmented audio chunks from the first Segmentation module are passed to the second Translation module, which leverages a powerful, instruction-tuned, off-the-shelf multimodal foundation model, Qwen3-Omni. Qwen3-Omni is a pretrained multilingual model based on a Mixture-of-Experts (MoE) architecture with state-of-the-art capabilities in cross-modal understanding and generation (Xu et al., 2025). Because of its native support for both text and audio modalities, proven strong offline speech recognition and translation performance, and ability to process long-form inputs, it is a suitable candidate for adaptation to the SimulST task.

Translation is performed in a *multi-turn conversation format*, where each turn includes a user message (audio chunk) and an assistant response (translation). The system enforces strict generation control with a variant of the wait-k read-write policy to regulate the lag between speech input and translation output, as well as repetition handling mechanisms. It also uses response prefilling and key-value caching to improve coherence and reduce redundant computation. These measures work together to deliver high-quality, incremental translations. This simple yet robust architecture enables real-time speech translation without the need for specialised training or adaptive modules.

2.1 Variable-Length Speech Segmentation

Incoming audio is segmented into variable-length chunks using the Silero VAD model (Silero Team, 2024), which estimates the speech probability of each 32-ms audio frame. Chunk boundaries are then determined with a simple hybrid approach that combines pause-based segmentation with fixed minimum and maximum chunk durations.

Audio accumulation is triggered when at least n_{a} frames within an n_{w}-frame window have speech probabilities greater than 0.5. Once the accumulated audio reaches the minimum chunk duration, the system searches for a natural pause and places the boundary at the start of the first frame whose speech probability falls below 0.5. If no such pause is detected before the accumulated audio reaches the maximum chunk duration, the boundary is instead placed at the frame with the lowest speech probability within the interval between the minimum and maximum durations. After a chunk is generated, audio accumulation is reset, and the system returns to the activation stage. This segmentation scheme aligns chunk boundaries with natural pauses whenever possible while limiting the latency caused by long stretches of continuous speech.

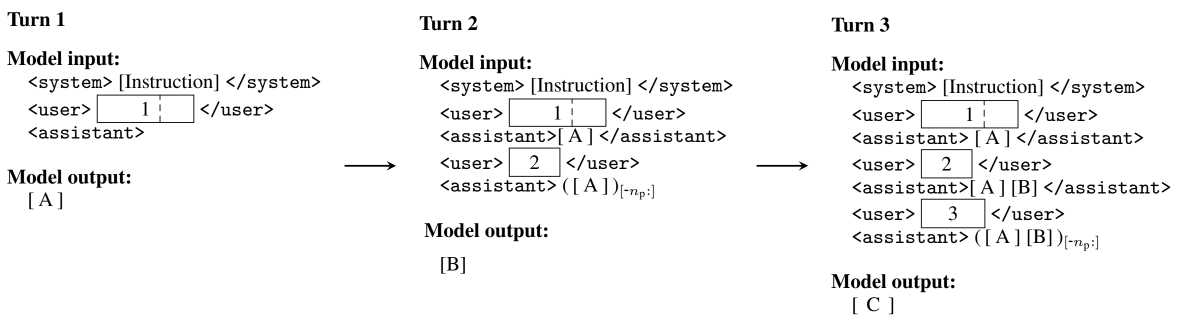

Figure 2: A conversation format with three user–assistant turns. Each turn consists of a user message containing a chunk of input audio (numbered rectangles) and an assistant translation response (square-bracketed capitals), whose prefix is forced to be the last n_p tokens of previous translations, and each turn's output can only have at most n units of text (excluding the prefix). A special case is the first turn, whose input consists of k audio chunks merged into one (in this figure, $k = 2$).

2.2 Adaptation for Speech Translation

Following Ouyang et al. (2025), we interface with the multimodal foundation model using a multi-turn conversation format. However, rather than fine-tuning the model to learn a read-write policy, we perform test-time adaptation through natural-language instructions and strict generation control. This enables the model to translate incoming speech incrementally while maintaining cross-turn coherence. The proposed test-time adaptation framework consists of four main components: a multi-turn conversation format, generation control with a read–write policy and repetition handling, as well as key-value caching for computational efficiency.

2.2.1 Multi-turn Conversation Format

The conversation format, as illustrated in Figure 2, consists of a system message containing an instruction and a sequence of *user–assistant* turns, where each user message is a chunk of input audio, and assistant responses are translation output. Below is the system instruction:

> You are a simultaneous speech translator. Translate the <source language> speech into <target language>. You will receive audio incrementally. Each response should be the *complete translation of all audio* received so far.

Crucially, the model is instructed to translate not only the latest chunk, but also previous ones, which is necessary because we control how much the model is allowed to generate in a turn (see Section 2.2.2), and there might be syntactical differences between the source and target languages. This instruction is coupled with *response prefilling* (Anthropic, 2026), a practice we have adapted from Koshkin et al. (2024), who implemented an LLM-based simultaneous text-to-text translator: the accumulated translation tokens from prior turns are injected as a prefix at the beginning of each assistant response after the first turn, so that generation resumes from exactly where the previous turn left off, ensuring coherence across turns. In order to reduce overhead, we prefill only a sliding window of the n_p most recent tokens rather than the entire translation history.

2.2.2 Read-Write Policy

We adapt the deterministic wait-(k, s, n) policy of Nguyen et al. (2021), in which the system waits for k acoustic frames before writing the first output token and subsequently alternates between reading s frames and writing n tokens. In our setting, where the input is segmented into variable-length audio chunks, the model first reads k chunks of audio. In each subsequent turn, it reads one additional chunk, corresponding to $s = 1$, and is allowed to generate at most n text units. The unit is word for English, German and Italian, and character for Chinese, excluding punctuation and space in both cases. The value of n is estimated based on the expected speaking rate and a language-direction-specific expansion factor, so that the generated output remains consistent with the desired read-write schedule.

In order to enforce this policy, we use a custom inference loop. At the end of the audio stream, the model is allowed to generate without restrictions until an end-of-sequence token is produced. Theoretically, in this policy, k controls how many chunks the translation is forced to lag behind the speech, and with $s = 1$ and the chunk duration configurable, the latter determines translation granularity. n should be set to an appropriate value according to the chunk duration so that the k-chunk lag holds. We find this policy to be a straightforward and easy-to-control mechanism to achieve a good trade-off between quality and latency, without the need for complex adaptive policies or training techniques.

2.2.3 Repetition Handling

An important part of our system is dedicated to preventing repetitions in the model's outputs and, if they still occur, detecting and discarding them. First, we penalise tokens already generated in a turn by subtracting their counts multiplied by a *penalty factor* α_{p} from their logits. We also check for literal repetitions of token n-grams at every decoding step: if there are more than two consecutive occurrences of an n-gram, we discard the last two and regenerate from the point where the first occurrence ends, at the same time setting the repeated tokens' logits to negative infinity to prevent them from being generated again. In addition to strict literal checks on tokens, we perform fuzzy matching on decoded text periodically and when repetitions are found, discard and regenerate them likewise.

2.2.4 Key-Value Caching

Since the conversation context grows monotonically as the audio stream progresses, we maintain the model's key-value (KV) cache throughout conversation turns. At each turn, only the new user message and the prefilled assistant response are processed, while the previous conversation history remains accessible through the cache. This avoids redundant computation and keeps the per-turn inference cost approximately proportional to the newly added audio input. Under the proposed multi-turn conversation format, KV caching allows the model to compute attention only over the latest turn's audio chunk and the assistant prefix, thereby substantially reducing computational cost.

3 Experiments

3.1 Experimental Datasets

We conduct experiments based on the development set for English-to-German/Chinese/Italian translation named MCIF (Papi et al., 2026), which consists of talks from the Association for Computational Linguistics (ACL) conferences, and that for Czech-to-English translation is a collection of recordings of the Chamber of Deputies' meetings in the Czech Parliament[1]. The statistics of the development sets are given in Table 1. It is worth mentioning that the recordings in the development set are substantially shorter than the maximum input audio duration supported by the Qwen3-Omni model (40 minutes). Therefore, we do not address support for infinite input streams in this work, although this can be easily done by evicting older conversation history from the model input.

	MCIF	**IWSLT26 Czech**
# Recordings	21	43
# Segments	919	679
Total Duration (min)	118	109
Total Segment Duration (min)	105	108
Average Segment Duration (sec.)	6.84	9.52
# Words	16,342	13,059
Average Words per Segment	17.8	19.2

Table 1: Statistics of the development sets.

3.2 Experimental Setup

3.2.1 Evaluation Metrics

Evaluation is conducted on unsegmented long-form audio, and the primary evaluation metrics are xCOMET (Guerreiro et al., 2024) for quality and LongYAAL (Polák et al., 2025) for latency. We target both latency regimes: **low latency** (0–2 seconds) and **high latency** (2–4 seconds).

3.2.2 Baseline Method

We compare our proposed system with the official baseline of IWSLT 2026[2], which is a cascaded system based on Qwen3-ASR-1.7B (Shi et al., 2026) for automatic speech recognition on fixed-size audio chunks and Qwen3-4B-Instruct-2507 (Yang et al., 2025) with the local agreement policy (Liu et al., 2020) for machine translation.

3.2.3 System Parameters

Parameter	**Symbol**	**Value**
Penalty factor	α_{p}	0.2
Assistant prefix window size	n_{p}	100
Activation window size	n_{w}	10
# activation frames	n_{a}	3
Min chunk duration (ms)	d_{min}	960
Max chunk duration (ms)	d_{max}	$[1600, 3520]$
# awaited chunks	k	$\{1, 2, 3\}$

Table 2: Main parameters of proposed SimulST system.

[1] http://ufallab.ms.mff.cuni.cz/~polak/iwslt26-cs-dev.zip

[2] https://github.com/owaski/iwslt-2026-baselines

The main parameters of our system are listed in Table 2, where some are fixed and others are given in ranges. In particular, d_{max}, k and n are to be determined by hyperparameter search. For d_{max}, we use a step size of 320 ms. As for n, it is related to the language direction and the chunk duration, and we determine a suitable range for each language direction based on the configured range of chunk duration, the source language's speaking rate and the direction's *expansion factor* (Ni et al., 2022): the expansion factor from Language A to B is the average ratio of the number of units in a piece of text in A to that in a translation in B. We assume a speaking rate of 100–150 words per minute for English (Barnard, 2022), and 94–143 for Czech (Psutka et al., 2003)[3], and the expansion factors are listed in Table 3. Then a range can be determined by multiplying the three. For hyperparameter search, we choose about 160 representative combinations instead of performing full grid search.

Direction	Expansion Factor
en $\rightarrow$ de	0.80–0.95
en $\rightarrow$ zh	1.50–1.80
en $\rightarrow$ it	1.15–1.25
cs $\rightarrow$ en	1.10–1.20

Table 3: Expansion factors for the four language translation directions (Trusted Translations, Inc., 2026).

We use the Qwen3-Omni-30B-A3B-Instruct[4] model as the multimodal foundation model.

3.3 Results and Analysis

3.3.1 Speech Segmentation Analysis

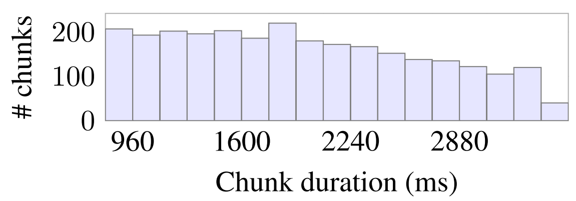

Figure 3: Distribution of chunk durations when $d_{\text{min}} = 960$ ms and $d_{\text{max}} = 3520$ ms on the MCIF dataset.

Our audio segmentation module is designed to generate chunks that respect natural speech boundaries whenever possible. To assess its effectiveness, we applied the module to the MCIF dataset and analyzed the resulting distribution of chunk durations, as shown in Figure 3. The results show that all chunk durations fall within the predefined minimum and maximum limits, while still exhibiting variable lengths. In addition, 79.8% of the chunks were segmented at frames where the speech probability was below 0.5, confirming the module's ability to align chunk boundaries with natural speech pauses.

3.3.2 Hyperparameter Search and Analysis

We conducted hyperparameter search on the development sets to find the best combinations of d_{max}, k and n in terms of quality–latency trade-off and generalisation ability. More specifically, as stated in Section 3.2.1, we evaluate quality with corpus-level COMET scores and latency with computation-unaware (CU) LongYAAL, and at the same time, we also compute recording-level COMET scores and use their standard deviation as a measure of generalisation ability. We use the standard deviation of COMET scores across the entire development set to provide a general indication of the proposed method's performance under the chosen hyperparameters, given constraints on time and computational resources. Cross-validation experiments are left for future work. After computing the metrics, we first divide the results under different settings into low- and high-latency regimes, then within each regime, we shortlist the top candidates that have the highest COMET scores. We look at all three metrics to decide the final setting for each language direction and latency regime, prioritising translation quality and generalisation ability over latency, as illustrated in Table 4.

As depicted in Figure 4, higher latency is generally correlated with better quality, except for some outliers with very small n or large d_{max}. This is because too small an n can 'suffocate' the model and make the translation fall far behind the speech's progress (the red/blue points in the bottom right corner of each subfigure of Figure 4). These hyperparameters themselves are also correlated—for example, a large d_{max} must be paired with an n large enough to make the model keep up with the speech, or the quality drops even with higher latency (green stars in Figure 4(b)); also, if d_{max} and n are in suitable ranges, a larger k can lead to better quality at the cost of higher latency. When all of the three hyperparameters are in appropriate ranges and well-coordinated, the system can achieve good quality with reasonable latency (points in the upper-left corner of each subfigure in Figure 4).

[3] In this study, we use the above range instead of 64–173 wpm in the original paper to speed up the search.

[4] https://huggingface.co/Qwen/Qwen3-Omni-30B-A3B-Instruct

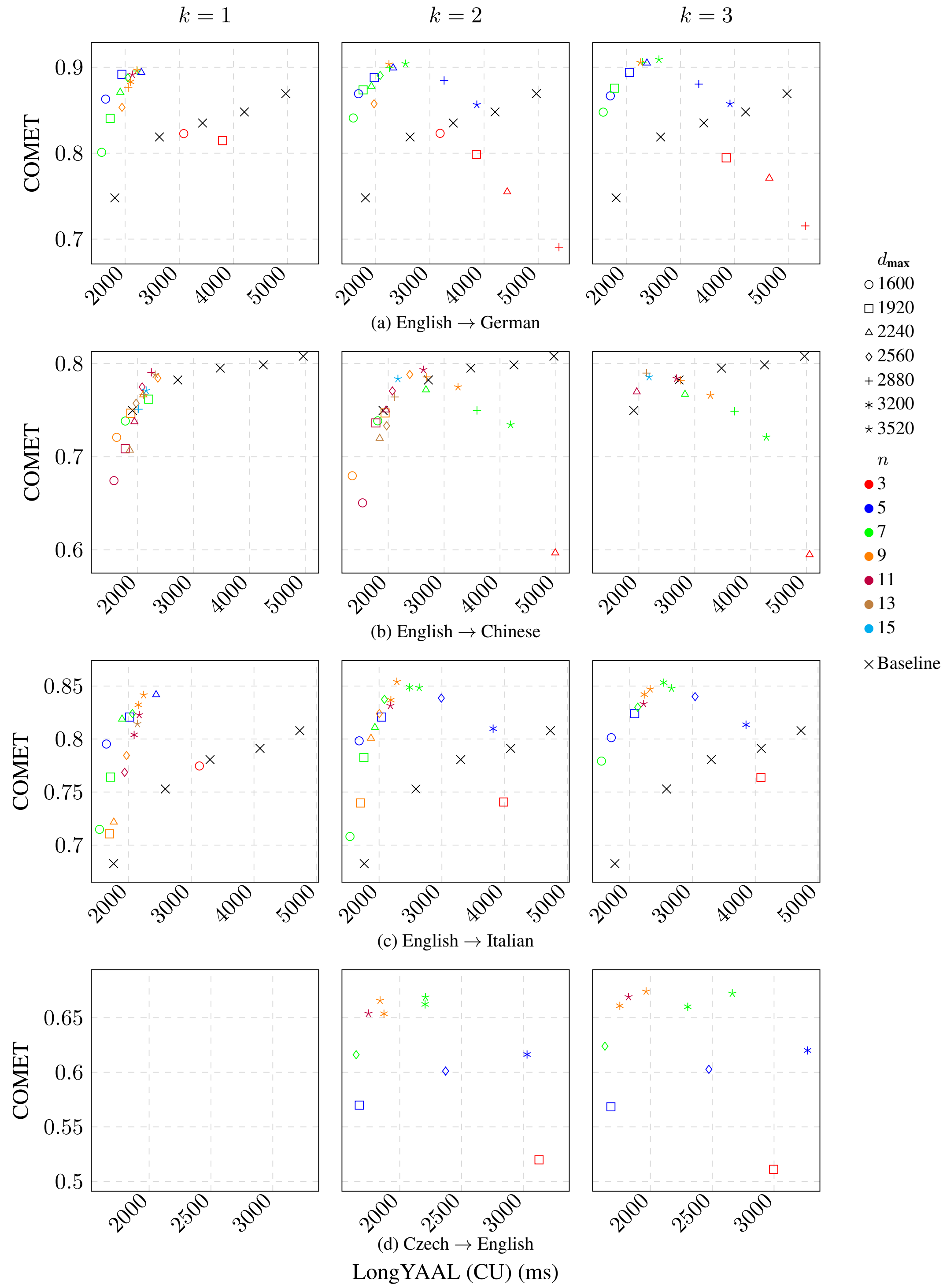

Figure 4: Quality–latency trade-off analysis for the four language directions under different hyperparameter combinations ($d_{\max}, k, n$). The hyperparameters should all be inside appropriate ranges to make the system achieve good performance; either too small or too large values can lead to suboptimal results. Note that the baseline results on (d) are not released.

3.3.3 Comparison with Baseline

As shown in Table 5, our system achieves a better overall quality–latency trade-off than the baseline on the MCIF dataset across the evaluated language directions, with particularly large gains for English-to-German and English-to-Italian translation, possibly because of the two pairs' genealogical affinity. For English-to-Chinese translation, the improvement is more modest: our system attains higher COMET in the low-latency setting and substantially lower latency in the high-latency setting, although

Direction	Latency	Setting			Metrics		
		d_{max}	k	n	COMET ↑	LongYAAL (CU) (ms) ↓	σ_{COMET} ↓
en → de	Low	1920	1	5	0.8918	1940	0.0330
		1920	2	5	0.8881	1972	0.0283
		2240	2	7	0.8779	1920	0.0469
	High	3520	3	7	0.9091	2601	0.0280
		2880	3	7	0.9060	2296	0.0256
		3520	3	9	0.9057	2258	0.0190
en → zh	Low	2240	3	11	0.7696	1962	0.0433
		2560	1	13	0.7574	1974	0.0559
		2240	2	11	0.7502	1961	0.0502
	High	3520	2	11	0.7934	2629	0.0352
		2880	1	11	0.7906	2247	0.0364
		2880	3	13	0.7898	2138	0.0344
en → it	Low	2240	1	7	0.8188	1897	0.0413
		2240	2	7	0.8109	1933	0.0631
		1600	3	5	0.8013	1706	0.0380
	High	3520	2	9	0.8541	2283	0.0347
		3200	3	7	0.8532	2542	0.0368
		3200	2	7	0.8487	2486	0.0392
cs → en	Low	3520	3	9	0.6742	1965	0.131
		3520	3	11	0.6691	1823	0.134
	High	3520	3	7	0.6723	2662	0.142
		3520	2	7	0.6688	2208	0.133

Table 4: Hyperparameter settings for each language direction and latency regime. The chosen settings of our submitted system are highlighted in blue.

Direction	Latency	System	COMET ↑	LongYAAL (CU) (ms) ↓
en → de	Low	Baseline	0.7480	**1810**
		Ours	**0.8881**	1972
	High	Baseline	0.8351	3431
		Ours	**0.9057**	**2258**
en → zh	Low	Baseline	0.7496	**1909**
		Ours	**0.7696**	1962
	High	Baseline	**0.7951**	3479
		Ours	0.7898	**2138**
en → it	Low	Baseline	0.6825	**1764**
		Ours	**0.8188**	1897
	High	Baseline	0.7806	3300
		Ours	**0.8541**	**2283**

Table 5: Performance comparison between the baseline method and our system on the MCIF dataset.

with a slight drop in COMET. This may reflect the inherent difficulty of translating in this language direction, as English and Chinese belong to distant language families and have considerable differences lexically and syntactically.

4 Conclusion

This paper presents our submission to the main track of the IWSLT 2026 SimulST shared task. Our system first applies a hybrid VAD-based approach for audio segmentation, and then performs translation with an offline multimodal foundation model adapted to the task at test time. For the test-time adaptation, we design a fixed read–write policy inspired by wait-k, together with a multi-turn conversation format that incorporates response prefilling and KV caching. We conduct a hyperparameter search over representative combinations of three key hyperparameters and select suitable configurations for each language direction and latency regime. Experiments on the official development sets show that our system outperforms the cascaded baseline for English-to-German, English-to-Chinese, and English-to-Italian translation, while also achieving reasonable zero-shot performance on Czech-to-English translation. These results demonstrate the effectiveness and efficiency of our approach.

Limitations

First of all, the multi-turn conversation format can lead to error accumulation: once a single turn goes awry, for example, by refusing to follow the instruction, subsequent turns might follow suit. Also, KV cache management can be further optimised to handle infinite audio streams by introducing a sliding window mechanism. In addition, hyperparameter search is coarse-grained and only performed on two small and single-domain datasets. Last but not least, inference costs of a large multimodal foundation model are significant because of its relatively large size, resulting in higher computation-aware latency.

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